

RABIN KHADKA

Data Science | Business Administration | Business Analytics | Supply Chain Decision-Making

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PROFESSIONAL SUMMARY

Data science professional with an M.S. in Data Science and a BBA, combining analytics experience in a logistics-focused business environment with prior classroom teaching. Research interests center on machine learning, statistical modeling, and explainable AI for business analytics and supply-chain decision-making.

PROFESSIONAL & TEACHING EXPERIENCE

Junior Social Media Data Analyst | Kaebox LLC — Winston-Salem, NC

April 2025–Present

- Build Python- and SQL-based workflows to ingest, clean, structure, and analyze digital engagement and text data for KPI monitoring and trend analysis.
- Apply regression, clustering, and classification methods to market and engagement data for consumer-behavior assessment and brand-sentiment analysis.
- Develop Tableau dashboards that translate social-media and logistics-related performance indicators into decision-oriented business insights.

Business Organization Instructor | Wits Academy — Boudha, Kathmandu, Nepal

2020–2022

- Taught Business Organization to Grade 11 and Grade 12 students, covering management, organizational structure, business operations, and decision-making.
- Prepared lessons, assessments, and instructional materials to support student learning and understanding of business concepts.
- Guided students in developing analytical thinking and applying business concepts to practical situations.

SELECTED RESEARCH & ANALYTICAL PROJECTS

Upstream Price Pressures and Consumer Inflation: Testing Supply Chain Price Transmission in the U.S. — Python | Time Series | Granger Causality

- Analyzed U.S. CPI, Producer Price Index, and Import Price Index data (BLS, Jan 2015–Jul 2026) using cross-correlation analysis and pairwise Granger-causality F-tests across multiple lag structures.
- Found import prices Granger-cause CPI changes ($p = 0.021$, one-month lag) with no reverse causality — evidence of a one-directional, exogenous upstream supply-chain shock. Producer and consumer prices showed strong bidirectional causality. Working paper. [GitHub](#)

Amazon Product Analytics: Sentiment Analysis & Predictive Modeling

Python | NLP | ML

- Analyzed 1,465 Amazon product records (16 original variables) to examine pricing, discounts, ratings, product categories, and customer-review sentiment through data cleaning and EDA.
- Applied TextBlob and VADER sentiment analysis to review titles and content, integrating unstructured customer feedback with structured product-level variables.
- Compared linear and ensemble regressors for discount prediction; tuned Gradient Boosting achieved the strongest test R^2 (0.3223). [GitHub](#)

Global Life Expectancy Prediction

Python | Regression | Random Forest

- Evaluated demographic, economic, and health indicators as predictors of life expectancy using regression, decision trees, and random forests; compared models using MAE, MSE, and R^2 .
- Random Forest produced the strongest predictive performance ($R^2 = 0.967$), substantially outperforming the tested regression and single-tree approaches; HIV/AIDS mortality and Income Composition of Resources accounted for ~77% of feature importance. [GitHub](#)

Insurance Fraud Detection Using Machine Learning

Python | Classification | SMOTE

- Prepared an imbalanced insurance-claims dataset for fraud classification using feature engineering, scaling, and SMOTE; compared Random Forest, Gradient Boosting, SVM, and neural-network approaches.
- Evaluated model generalization with train/test metrics and cross-validation, emphasizing class imbalance, overfitting, and careful model selection in fraud-risk prediction.

EDUCATION

Carolina University — Winston-Salem, NC

May 2024

Master of Science in Data Science | GPA: 3.83/4.00

Selected Coursework: Statistical Modeling; Big Data Analytics; Machine Learning Algorithms; Advanced Database Management Systems

TECHNICAL & QUANTITATIVE SKILLS

Programming & Data: Python, R, SQL, pandas, NumPy, scikit-learn, statsmodels

Machine Learning & Statistics: Regression, classification, random forests, gradient boosting, SVM, Naive Bayes, PCA, cross-validation, hyperparameter tuning, Granger causality

Deep Learning & Text Analytics: TensorFlow/Keras, multilayer perceptrons, NLP, sentiment analysis, TextBlob, NLTK/VADER

Visualization & Data Systems: Tableau, Matplotlib, Seaborn, database management, API data extraction

ACADEMIC & CAREER OBJECTIVE

Pursue doctoral research at the intersection of business analytics, supply-chain decision-making, and trustworthy AI, with emphasis on procurement and supplier management, demand forecasting, inventory and warehouse operations, and explainable machine learning. Long-term goals include a faculty career at a research university and complementary work in industry R&D.